UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the Interstate Transmission System	))))	Docket No. RM26-4

INITIAL COMMENTS OF MICROSOFT, INC.

Pursuant to the October 27, 2025 Notice Inviting Comments of the Federal Energy Regulatory Commission ("Commission") and the November 7, 2025 Notice Granting Extension of Time, Microsoft Corporation ("Microsoft") comments on the Advanced Notice of Proposed Rulemaking ("ANOPR"). The ANOPR invites comment on 14 framework principles proposed to inform the Commission's rulemaking deliberations regarding the interconnection of large loads and hybrid facilities to jurisdictional transmission facilities.

The U.S. economy relies on an electric grid that, at present, cannot meet the rapidly increasing electricity requirements of the digital economy and datacenters, reshoring of manufacturing and heavy industry, and electrification.[1] Microsoft recognizes that the expansion of data centers performing cloud computing and artificial intelligence ("AI") operations necessitates significant investment in electric transmission and generation capacity, support for domestic supply chains, and new tools and technologies that optimize grid operations. Microsoft is investing heavily in the electric power sector and appreciates the initiative of the U.S. Department of Energy ("DOE") and the Commission to streamline, standardize, modernize, and speed up access to the electric grid for datacenters and other

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[1] *See* Brad Smith, *Winning the AI race: Strengthening U.S. capabilities in computing and innovation*, MICROSOFT ON THE ISSUES (May 8, 2025), https://blogs.microsoft.com/on-the-issues/2025/05/08/winning-the-ai-race/.

large loads, and for inviting industry participants to address the pressing challenges of grid interconnection and service to new large loads and hybrid facilities. Together, the administration, industry, customers, and stakeholders can develop policies that speed up access to power to deliver economic growth and ensure energy affordability and reliability for all Americans.

Microsoft's comments focus on five ANOPR principles necessary for the U.S. datacenter industry: integrated studies of large loads, hybrid facilities, and generator-only interconnections to achieve reliable, efficient, and cost-effective interconnection to the electric grid (**principle no. 3**); standardized study processes that remove uncertainty for customers, utilities, and entities responsible for regional grid operations and planning (**principle no. 4**); expedited study processes for large loads that agree to voluntarily curtail during system stress and hybrid facilities that likewise agree to be curtailable and dispatchable (**principle no. 7**); cost allocation rules that fairly and equitably allocate costs among system users (**principle no. 8**); and transition mechanisms that avoid regulatory uncertainty (**principle no. 13**). These ANOPR principles present an important opportunity for collaboration among state and federal regulators to develop an open and nondiscriminatory framework for interconnecting large loads and hybrid facilities that facilitates coordination, provides transparency for system users, achieves efficiencies, lowers costs, and enhances regulatory certainty.

I. BACKGROUND

Microsoft's datacenters deliver world-class data security and privacy, enabling the applications, capabilities, and services that support the rapidly evolving digital economy.[2] Microsoft operates 24/7 to support critical infrastructure and services—e.g., first responders, hospitals, and emergency services; financial institutions; services across the food and agricultural distribution supply chain; and federal, state, and local government operations essential for public health, transportation, and national security.[3] Indeed, the services that Microsoft provides are integral to the U.S. economy—Microsoft Azure is used by 95% of Fortune 500 companies and deployment of generative AI could potentially realize $3.8 trillion in productivity savings by 2038.[4] Microsoft's datacenters also make possible everyday activities like paying household bills, remote work, and video calls among family and friends.

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[2] *See Microsoft Datacenters: Illuminating the unseen power of the cloud*, MICROSOFT, https://datacenters.microsoft.com/ (last visited Nov. 18, 2025).

[3] *See Understanding Microsoft Datacenters,* MICROSOFT, https://news.microsoft.com/datacenters/#:~:text=Microsoft%20datacenters%20play%20a%20crucial,miles%20of%20fiber%20optic%20cables (last visited Nov. 18, 2025); Bill Briggs, *'Critical to our modern society': How datacenters power everyday necessities*, MICROSOFT (Oct. 26, 2022), https://news.microsoft.com/source/emea/features/critical-to-our-modern-society-how-datacenters-power-everyday-necessities/.

[4] *See Microsoft Azure: The only consistent, comprehensive hybrid cloud*, MICROSOFT AZURE BLOG (Sept. 25, 2018), https://azure.microsoft.com/en-us/blog/microsoft-azure-the-only-consistent-comprehensive-hybrid-cloud/?msockid=2eef0b747bf0658d126f189d7aa66484; Rima Alaily, *The $3.8 trillion opportunity: unlocking the economic potential of the US generative AI ecosystem*, MICROSOFT ON THE ISSUES (Nov. 21, 2024), https://blogs.microsoft.com/on-the-issues/2024/11/21/the-3-8-trillion-opportunity-unlocking-the-economic-potential-of-the-us-generative-ai-ecosystem/.

Microsoft constructs, owns, and operates more than 400 datacenters in 70 regions across North America and five continents, more than any other cloud provider.[5] Microsoft's investments are guided by the needs of its business and government customers that use its services. Microsoft's customers can choose the datacenter regions where computing power and services are deployed for them based on needs including network performance and data residency. Many of Microsoft's customers require that their data be stored in the United States, so it is critical that Microsoft be positioned to expand operations to continuously meet its obligations to its customers and contribute to the United States' leadership in AI.

Microsoft has datacenters operating or under development across the U.S., including in Arizona, California, Georgia, Illinois, Iowa, Texas, Virginia, Washington, Wisconsin, and Wyoming. This year, Microsoft announced two major AI datacenter facilities: the first, in southeastern Wisconsin, is expected to come online in early 2026, and the second, in Atlanta, Georgia, began operation in October 2025.[6] These facilities are the first phase of a dedicated network of sites that will operate as an AI superfactory to accelerate AI breakthroughs and train new models on a scale that, until now, has been impossible.[7]

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[5] *See Azure global infrastructure: Grow with confidence and choice on a trusted, sustainable, and advanced cloud and AI infrastructure*, MICROSOFT, https://azure.microsoft.com/en-us/explore/global-infrastructure#:~:text=Azure%20global%20infrastructure (last visited Nov. 18, 2025). Microsoft's latest quarterly earnings report datacenter investments of $24 billion globally and this pace is expected to continue. *See Press Release & Webcast: Earnings Release FY26 Q1*, Microsoft (Oct. 29, 2025), https://www.microsoft.com/en-us/investor/earnings/FY-2026-Q1/press-release-webcast.

[6] *See, e.g.,* Catherine Bolgar, *From Wisconsin to Atlanta: Microsoft connects datacenters to build its first AI superfactory,* MICROSOFT (Nov. 12, 2025), https://news.microsoft.com/source/features/ai/from-wisconsin-to-atlanta-microsoft-connects-datacenters-to-build-its-first-ai-superfactory/.

[7] *Id.*

Microsoft is actively engaged in discussions with national, state, and local stakeholders on energy infrastructure, permitting, and community partnerships to support the responsible growth of datacenters. Locating datacenters in the United States helps ensure the country remains at the forefront of innovation, strengthens supply chain security, and delivers economic benefits to local communities. Through its Datacenter Community Pledge, Microsoft commits that datacenters will not only serve as the backbone of modern technology but also provide benefits to the communities where they are located.[8] As a large electricity consumer, Microsoft pays its share of electric system costs and is committed to working closely with regulators, grid operators, transmission-owning utilities, load serving entities, power producers, and other stakeholders with the objective of ensuring that U.S. electricity infrastructure is reliable, resilient, and affordable to all.

Microsoft expects that consumer-driven demand for cloud services and AI applications and technologies will accelerate rapidly over at least the next decade. Timely access to electric power supplies has been and continues to be the biggest obstacle industry-wide to the deployment of advanced computing technology in the United States. But it is not only about timely access. Microsoft requires reliable and resilient electric power supplies to assure the reliability of its datacenters because Microsoft's datacenters support critical infrastructure services essential for the welfare of the United States' economy and its people.[9]

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[8] Noelle Walsh, *Microsoft's Datacenter Community Pledge: To build and operate digital infrastructure that addresses societal challenges and creates benefits for communities*, OFFICIAL MICROSOFT BLOG (June 2, 2024), https://blogs.microsoft.com/blog/2024/06/02/microsofts-datacenter-community-pledge-to-build-and-operate-digital-infrastructure-that-addresses-societal-challenges-and-creates-benefits-for-communities/.

[9] Microsoft commits to service availability levels of up to 99.99% of the time, and power system reliability and power redundancy play a large part in ensuring Microsoft can

Microsoft is already substantially invested in the power sector. Last year, Microsoft announced a long-term, front-of-the-meter power purchase agreement with Constellation to restart the Unit 1 reactor at the Crane Energy Center in Dauphin County, Pennsylvania.[10] This restart is expected by 2028, adding more than 800 megawatts of new firm, carbon-free energy and electric capacity to the electric grid operated by PJM Interconnection, L.L.C.[11] Additionally, Microsoft is investing in projects and companies at the cutting edge of advanced energy system technology. Two years ago, Microsoft executed a power purchase agreement for offtake with the first nuclear fusion generator, Helion Energy.[12] Further, through its $1 billion Climate Innovation Fund, Microsoft is providing innovative financing

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meet its customer commitments. *See Reliability with Microsoft Azure*, MICROSOFT AZURE, https://azure.microsoft.com/files/Features/Reliability/AzureResiliencyInfographic.pdf?v=95f7f9240e31cb9d723ea0cfdea7864bef338788e9324919e9a93635fb8f64c5&msockid=2eef0b747bf0658d126f189d7aa66484 (last visited Nov. 18, 2025).

[10] Bobby Hollis, *Accelerating the addition of carbon-free energy: An update on progress,* MICROSOFT (Sept. 20, 2024), https://www.microsoft.com/en-us/microsoft-cloud/blog/2024/09/20/accelerating-the-addition-of-carbon-free-energy-an-update-on-progress/ (linking *Constellation to Launch Crane Clean Energy Center, Restoring Jobs and Carbon-Free Power to the Grid*, CONSTELLATION (Sept. 20, 2024), https://www.constellationenergy.com/newsroom/2024/Constellation-to-Launch-Crane-Clean-Energy-Center-Restoring-Jobs-and-Carbon-Free-Power-to-The-Grid.html).

[11] *Constellation to Launch Crane Clean Energy Center, Restoring Jobs and Carbon-Free Power to the Grid*, CONSTELLATION (Sept. 20, 2024), https://www.constellationenergy.com/newsroom/2024/Constellation-to-Launch-Crane-Clean-Energy-Center-Restoring-Jobs-and-Carbon-Free-Power-to-The-Grid.html.

[12] Michelle Patron, *Advocating for decarbonization of the power sector*, MICROSOFT ON THE ISSUES (Aug. 16, 2023), https://blogs.microsoft.com/on-the-issues/2023/08/16/apec-sustainability-decarbonization-fusion-energy/ (linking Timothy Gardner, *Microsoft signs power purchase deal with nuclear fusion company Helion*, REUTERS (May 10, 2023), https://www.reuters.com/technology/microsoft-buy-power-nuclear-fusion-company-helion-2023-05-10/).

to companies such as LineVision, AutoGrid, and Utilidata.[13] Microsoft expects its investments will contribute to the development of a stronger and more robust electric grid.

Microsoft appreciates consideration of policies by DOE, the Commission, public utility commissions, and state and local energy departments that will position grid operators and electric utilities to reliably interconnect and serve datacenters and other large commercial and industrial retail loads. That need is happening now and is expected to continue at this pace across the United States through at least the next decade.

II. COMMENTS

On October 23, 2025, Secretary of Energy Chris Wright directed the Commission to initiate proposed rulemaking procedures and to consider the ANOPR's potential reforms intended to ensure the timely and orderly interconnection of large loads to Commission-jurisdictional transmission facilities.[14] On October 27, 2025, and as amended by the November 7, 2025 notice granting an extension of time, the Commission issued a notice docketing the ANOPR, inviting comments, and setting the date for initial and reply comments.[15] The ANOPR seeks to engage industry stakeholders by proposing interconnection procedures for large loads (greater than 20 MW) and hybrid facilities, which

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[13] *Climate Innovation Fund*, MICROSOFT, https://www.microsoft.com/en-us/corporate-responsibility/sustainability/climate-innovation-fund (last visited Nov. 18, 2025).

[14] *See Secretary of Energy's Direction that the Federal Energy Regulatory Commission Initiate Rulemaking Procedures and Proposal Regarding the Interconnection of Large Loads Pursuant to the Secretary's Authority Under Section 403 of the Department of Energy Organization Act* (Oct. 23, 2025) ("Secretary Wright's Letter").

[15] *See Interconnection of Large Loads to the Interstate Transmission System,* Notice Inviting Comments, Docket No. RM26-4-000 (Oct. 27, 2025); *Interconnection of Large Loads to the Interstate Transmission System,* Notice Granting Extension of Time, Docket No. RM26-4-000 (Nov. 7, 2025).

are large loads co-located with, electrically adjacent to, or that share a point of interconnection with new or existing generation facilities, and that directly interconnect to interstate transmission facilities subject to the Commission's jurisdiction.[16] The ANOPR asks for comment on framework principles that may inform the Commission's rulemaking deliberations regarding interconnection of these large loads and hybrid facilities.

In consideration of the urgent need for action, the Commission is expected to take final action by no later than April 20, 2026.[17] As explained in the ANOPR, electricity demand in the United States is accelerating and load growth is greater than at any point in recent history. Microsoft understands that the interconnection of datacenters and other large loads presents new challenges for state and federal utility regulators, utilities, and grid operators. Microsoft supports collaboration and coordination among state and federal regulators for interconnecting large loads and hybrid facilities and respectfully requests that the Commission take these comments into consideration as it decides its next steps.

A. Principle #3: Combined Generation and Load Studies

The ANOPR's third principle proposes, to the extent possible, load and hybrid facilities should be studied together with generating facilities. For example, the ANOPR proposes that siting a large "baseload" load near or at the same point of interconnection as a new generating facility (one possible form of bring-your-own generation or "BYOG") could reduce the number and scope of network upgrades necessary for both the load and generator interconnections. Microsoft believes studying load and generation together has the potential to speed up the interconnection process for load and generation by eliminating duplicative,

[16] *See* ANOPR at PP 12, 19.

[17] Secretary Wright's Letter at 2.

and potentially conflicting, studies and will facilitate more efficient grid planning by more closely aligning generation capacity needs with large loads. Additionally, studying load and generation together potentially reduces the scale and cost of transmission upgrades by minimizing system impacts and the need for major backbone reinforcements. Eliminating unnecessary regulatory and interconnection delays—while assuring system reliability—is necessary to match the pace of AI innovation in the United States. In Microsoft's view, the changes ripe for consideration are those that (i) focus on processes that will increase speed-to-market for new customers and new generation and (ii) reduce the cost and scope of network upgrades needed for interconnection.

B. Principle #4: Standard Study Requirements

The ANOPR's fourth principle proposes that load and hybrid facilities be subject to standardized study deposits, readiness requirements, and withdrawal penalties.[18] The ANOPR presents these reforms not only to deter speculative projects, but also to provide entities with more reliable information that they can use with confidence in system planning activities. Microsoft supports standardizing study requirements to provide greater regulatory certainty, deter speculation, and enhance transparency so businesses can plan with more predictability. Interconnection processes should clearly articulate study requirements, financial commitments, timelines, and responsibilities of the utility and the customer. It is important to Microsoft to receive studies that are timely delivered and provide robust results and reasonably reliable cost estimates.

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[18] ANOPR at P 21. As noted above, Microsoft takes no position on whether the Commission should exercise jurisdiction over load interconnections.

An additional benefit of interconnection study reform is that it could bring greater consistency to system-level load forecasting for system planners. Such a load forecasting process could provide greater transparency and accuracy by creating a feedback loop that allows large loads to validate the load levels utilities and regional planning entities are incorporating into their forecasts.

C. Principle #7: Voluntarily Curtailable Loads

The ANOPR proposes an expedited study process for the interconnection of large loads that voluntarily agree to be curtailable and hybrid facilities that agree to be curtailable and dispatchable.[19] Microsoft agrees that regulatory frameworks that allow for voluntarily curtailable loads make sense and will be an important tool for accelerating the deployment of critical AI infrastructure. As discussed briefly below, most data centers today have limited curtailment capability. However, a regulatory framework that provides accelerated interconnection processes and clear market signals for curtailable loads should incentivize innovation in datacenter design and operations. Further, flexibility and demand response programs can unlock grid benefits, including relieving system constraints, enabling efficient use of existing capacity, and providing a bridge-to-power while the generation and transmission infrastructure expansion necessary for long-term firm reliable supplies is designed, permitted, and constructed. Microsoft supports consideration of reforms that enable utilities, power companies, and consumers to deploy grid-flexible technologies.

Microsoft believes an expedited study process for curtailable loads and dispatchable hybrid facilities would be appropriate and consistent with the interests of the administration, so long as these arrangements do not create operational reliability risks and do not interfere

[19] *Id*. at P 25.

with the interconnection of critical nonflexible large loads. Technological and new market mechanisms that might be included in an expedited study process could include incremental datacenter operator-funded generation or storage and innovative load management capabilities. Microsoft's data centers primarily host cloud services for its customers, including government and business customers who require service levels and data residency commitments that are not compatible with curtailing or shifting cloud computing operations from one datacenter to another.

Microsoft encourages regulators, utilities, and grid operators—with consultation from datacenter operators—to create program clarity by identifying the type of flexibility that would bring value to the electric system and developing program parameters (such as maximum load curtailment times, frequency, and size) that are critical for datacenters to consider when making decisions about program participation.

D. Principle #8: Direct Assignment of Network Costs

The ANOPR proposes that large loads and hybrid facilities be responsible for 100% of the upfront costs of assigned network upgrades.[20] Whatever the mechanism by which large loads or their load serving entities are directly assigned network upgrade costs, Microsoft recognizes its responsibility as a large retail electricity consumer to ensure the electric grid is reliable, resilient, affordable, and accessible for all. Today, Microsoft is already deploying billions of dollars to support the energy infrastructure build-out necessary to grow its business. That investment will continue as demand for Microsoft's Azure cloud and AI services grows. More broadly, Microsoft is interested in innovative solutions that help to accelerate much-needed investment in the transmission system.

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[20] *Id*.

Importantly, Microsoft pays for the electricity it consumes and for its share of infrastructure costs to generate and deliver electricity to datacenter sites. Microsoft supports the work of utilities and regulators to make sure these costs are transparently calculated and assigned to it and the other customers who benefit from those capital investments. This work helps ensure that Microsoft's neighbors and the local community do not pay for Microsoft's share of the costs.

The ANOPR requests comment on whether network upgrade costs paid up-front by a large load should be offset through a crediting mechanism. If the network upgrades (with costs paid for up-front by the interconnecting loads they are assigned to) are providing broad benefits to other system users—for example, if another industrial customer benefits from the same infrastructure —Microsoft believes it would be fair to use a crediting mechanism or other arrangement that would capture the reliability and economic value those network upgrades provide other electricity consumers. One such approach may be similar to the crediting mechanisms used for interconnecting generator-only facilities.

E. Principle #13: Transition Plan

The ANOPR proposes that transition plans be developed for the treatment of large load interconnections that are already being studied for interconnection. Microsoft is supportive of transition plans that mitigate existing risk and do not create new regulatory risk or uncertainty for projects. A transition plan should provide regulatory stability for large loads that are already under study; large loads for which the study process has already been completed, payments made, and construction commenced; and for facilities that are phasing in or are fully operational.

III. CONCLUSION

Microsoft appreciates DOE's and the Commission's leadership in seeking innovative solutions to the challenges of interconnection and providing electric service to new large loads and hybrid facilities. Microsoft respectfully requests that the Commission consider its comments in developing policies to support the availability of reliable, secure energy supplies to meet the needs of its datacenters and all system users.

Respectfully submitted,

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